

PROJECT FILE

WORKING WITH MONGO DB DOCUMENTS AND COLLECTIONS

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CREATE OPERATION- The **Create** operation is used to add new documents to a collection. MongoDB provides methods like insertOne() for single documents and insertMany() for multiple documents.

Read Operation- The Read operation retrieves documents from a collection. MongoDB uses methods like find() to fetch multiple documents and findOne() to retrieve a single document.

```
>_MONGOSH
> use("CollegeManagement")
< switched to db CollegeManagement
> db["student"].find()
<
> db.students.insertOne({
  roll: 1,
  name: "Aditi",
  age: 19,
  marks: 85,
  city: "Lucknow"
})
< {
  acknowledged: true,
  insertedId: ObjectId('6ab65076022c9fef69f06336')
}
```

```
_id: ObjectId('6ab65076022c9fef69f06336'),
roll: 1,
name: 'Aditi',
age: 19,
marks: 85,
city: 'Lucknow'
}
```

```
> db.students.insertMany([
  {
    roll: 2,
    name: "Rohan",
    age: 20,
    marks: 78,
    city: "Kanpur"
  },
  {
    roll: 3,
    name: "Mehak",
    age: 18,
    marks: 92,
    city: "Delhi"
  },
  {
    roll: 4,
    name: "Arjun",
    age: 19,
    marks: 74,
    city: "Jaipur"
  },
  {
```

```
    roll: 5,
    name: "Simran",
    age: 20,
    marks: 88,
    city: "Mumbai"
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab653f4022c9fef69f06337'),
    '1': ObjectId('6ab653f4022c9fef69f06338'),
    '2': ObjectId('6ab653f4022c9fef69f06339'),
    '3': ObjectId('6ab653f4022c9fef69f0633a')
  }
}
```

```
> db.students.find({
  roll: {$in: [1,2, 3, 4, 5]}
})
< {
  _id: ObjectId('6ab606348d52530e0c393979'),
  roll: 1,
  name: 'Aditya',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab606348d52530e0c39397a'),
  roll: 2,
  name: 'Sneha',
  age: 19,
  marks: 89,
  city: 'Mumbai'
}
{
  _id: ObjectId('6ab606348d52530e0c39397b'),
  roll: 3,
```

```
{
  _id: ObjectId('6ab606348d52530e0c39397b'),
  roll: 3,
  name: 'Karan',
  age: 20,
  marks: 94,
  city: 'Jaipur'
}
{
  _id: ObjectId('6ab606348d52530e0c39397c'),
  roll: 4,
  name: 'Priya',
  age: 18,
  marks: 65,
  city: 'Agra'
}
{
  _id: ObjectId('6ab606348d52530e0c39397d'),
  roll: 5,
  name: 'Rahul',
  age: 19,
  marks: 95,
```

```
{
  _id: ObjectId('6ab606348d52530e0c39397d'),
  roll: 5,
  name: 'Rahul',
  age: 19,
  marks: 95,
  city: 'Delhi'
}
```

UPDATE OPERATION- The Update operation modifies existing documents. MongoDB provides methods like update One() to update a single document and update Many() to update multiple documents. The \$set operator is commonly used to specify the fields to update.

```
> db.students.updateOne(  
  {roll: 2},  
  {$set: {marks: 90}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
{  
  _id: ObjectId('6ab606348d52530e0c39397a'),  
  roll: 2,  
  name: 'Sneha',  
  age: 19,  
  marks: 90,  
  city: 'Mumbai'  
}
```

DELETE OPERATION- The Delete operation removes documents from a collection. MongoDB offers deleteOne() to delete a single document and deleteMany() to remove multiple documents based on a filter.

```
> db.students.deleteMany({
  roll: {$in: [2, 3, 4]}
})
< {
  acknowledged: true,
  deletedCount: 6
}
```

```
> db.students.find({
  roll: {$in: [2, 3, 4]}
})
<
CollegeManagement>
```

```
> db.students.deleteOne({roll: 10})
< {
  acknowledged: true,
  deletedCount: 1
}
```



```
{
  _id: ObjectId('6ab653f4022c9fef69f06337'),
  roll: 2,
  name: 'Rohan',
  age: 20,
  marks: 78,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab653f4022c9fef69f06338'),
  roll: 3,
  name: 'Mehak',
  age: 18,
  marks: 80,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab653f4022c9fef69f06339'),
  roll: 4,
  name: 'Arjun',
  age: 19,
  marks: 74,
  city: 'Jaipur'
```

Using greater than in Mongo DB

```
> db.students.find({marks: {$gt: 85}})
< {
  _id: ObjectId('6ab606348d52530e0c39397a'),
  roll: 2,
  name: 'Sneha',
  age: 19,
  marks: 89,
  city: 'Mumbai'
}
{
  _id: ObjectId('6ab606348d52530e0c39397b'),
  roll: 3,
  name: 'Karan',
  age: 20,
  marks: 94,
  city: 'Jaipur'
}
```

Using less than in Mongo DB

```
> db.students.find({marks: {$lt: 60}})
< {
  _id: ObjectId('6ab606348d52530e0c393981'),
  roll: 9,
  name: 'Vansh',
  age: 20,
  marks: 57,
  city: 'Jaipur'
}
{
  _id: ObjectId('6ab606348d52530e0c39398a'),
  roll: 18,
  name: 'Kriti',
  age: 20,
  marks: 59,
  city: 'Agra'
}
```

Using greater than or equal to in Mongo DB- \$gte

```
db.students.find({marks: {$gte: 90}})

{
  _id: ObjectId('6ab606348d52530e0c39397b'),
  roll: 3,
  name: 'Karan',
  age: 20,
  marks: 94,
  city: 'Jaipur'
}
{
  _id: ObjectId('6ab606348d52530e0c39397e'),
  roll: 6,
  name: 'Ananya',
  age: 20,
  marks: 91,
  city: 'Pune'
}
```

Using less than or equal to in Mongo DB- \$lte

```
> db.students.find({marks: {$lte: 70}})
< {
  _id: ObjectId('6ab606348d52530e0c39397c'),
  roll: 4,
  name: 'Priya',
  age: 18,
  marks: 65,
  city: 'Agra'
}
{
  _id: ObjectId('6ab606348d52530e0c393981'),
  roll: 9,
  name: 'Vansh',
  age: 20,
  marks: 57,
  city: 'Jaipur'
}
```

Using Conditional Operation in Mongo DB

AND Operation:-

```
> db.students.find({
  $and: [
    {marks: {$gt: 80}},
    {age: 19}
  ]
})
< {
  _id: ObjectId('6ab606348d52530e0c39397a'),
  roll: 2,
  name: 'Sneha',
  age: 19,
  marks: 89,
  city: 'Mumbai'
}
{
  _id: ObjectId('6ab606348d52530e0c39397d'),
  roll: 5,
  name: 'Rahul',
  age: 19,
  marks: 95,
  city: 'Delhi'
}
```

Using Conditional Operation in Mongo DB

OR Operation:-

```
> db.students.find({
  $or: [
    {city: "Delhi"},
    {city: "Mumbai"}
  ]
})
< {
  _id: ObjectId('6ab606348d52530e0c393979'),
  roll: 1,
  name: 'Aditya',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab606348d52530e0c39397a'),
  roll: 2,
  name: 'Sneha',
  age: 19,
  marks: 89,
  city: 'Mumbai'
}
```

Using Sort Operation in Mongo DB-


```
db.students.find().sort({marks: 1})
```

```
{  
  _id: ObjectId('6ab606348d52530e0c393994'),  
  roll: 28,  
  name: 'Manish',  
  age: 18,  
  marks: 61,  
  city: 'Mumbai'  
}  
  
{  
  _id: ObjectId('6ab606348d52530e0c3939a9'),  
  roll: 49,  
  name: 'Ayesha',  
  age: 18,  
  marks: 62,  
  city: 'Pune'  
}
```



```
> db.students.find().sort({marks: -1})
```

```
< {  
  _id: ObjectId('6ab606348d52530e0c39399b'),  
  roll: 35,  
  name: 'Aarushi',  
  age: 19,  
  marks: 96,  
  city: 'Delhi'  
}  
  
{  
  _id: ObjectId('6ab606348d52530e0c39397d'),  
  roll: 5,  
  name: 'Rahul',  
  age: 19,  
  marks: 95,  
  city: 'Delhi'  
}
```

Using Limit Operation in Mongo DB-

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6ab606348d52530e0c393979'),
  roll: 1,
  name: 'Aditya',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab606348d52530e0c39397a'),
  roll: 2,
  name: 'Sneha',
  age: 19,
  marks: 89,
  city: 'Mumbai'
}
```

Using Search Operation in Mongo DB-

SEARCH by City

```
> db.students.find({city: "Delhi"})
< {
  _id: ObjectId('6ab606348d52530e0c393979'),
  roll: 1,
  name: 'Aditya',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab606348d52530e0c39397d'),
  roll: 5,
  name: 'Rahul',
  age: 19,
  marks: 95,
  city: 'Delhi'
}
```

COUNT DOCUMENTS-

```
> db.students.countDocuments()  
< 44  
CollegeManagement > |
```